

SMA-1 — All Paper Figures

Contact sheet for inspection — one figure per page

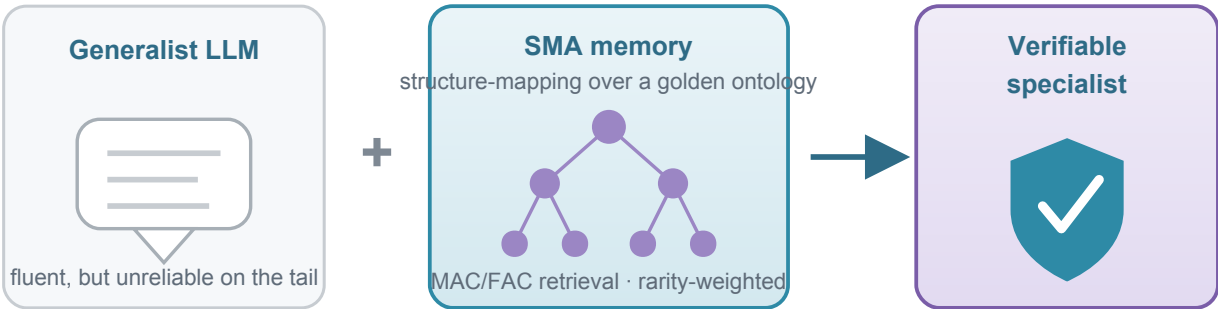
Main = in the compiled manuscript ED = Extended Data / supplement Alt = produced, not currently placed

June 13, 2026

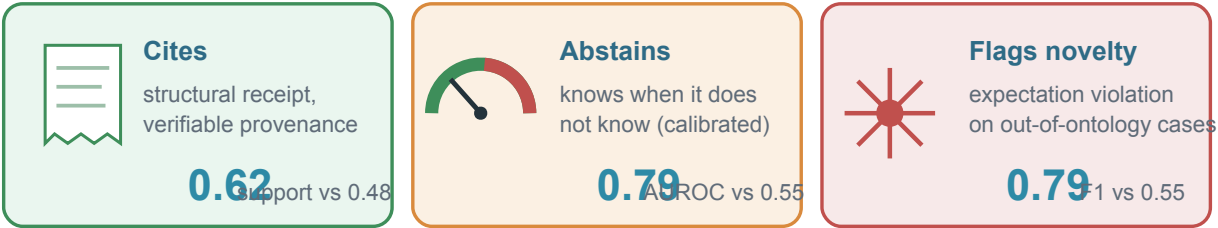
Graphical Abstract

Alt — single-panel one-shot story (cover candidate; not in manuscript body)

A golden-ontology structure-mapping memory turns a generalist LLM into a verifiable specialist



Three capabilities vector RAG and knowledge graphs structurally lack



Medicine LLM-QA, LLM & prompt fixed, memory swapped — SMA vs dense RAG: **accuracy 0.342 vs 0.100**
Generalizes across 5 domains: tail top-5 wins (med +0.33, gen +0.16, fin +0.17, cyber +0.07, legal +0.06; all Holm-sig.)

Figure 1 — Universal-adaptor system overview

MAIN (in manuscript) — (a) mount ontology as geometry (b) what each retriever keeps/discards (c) MAC→FAC→cite + abstain/novelty (d) router + tail-top5 radar

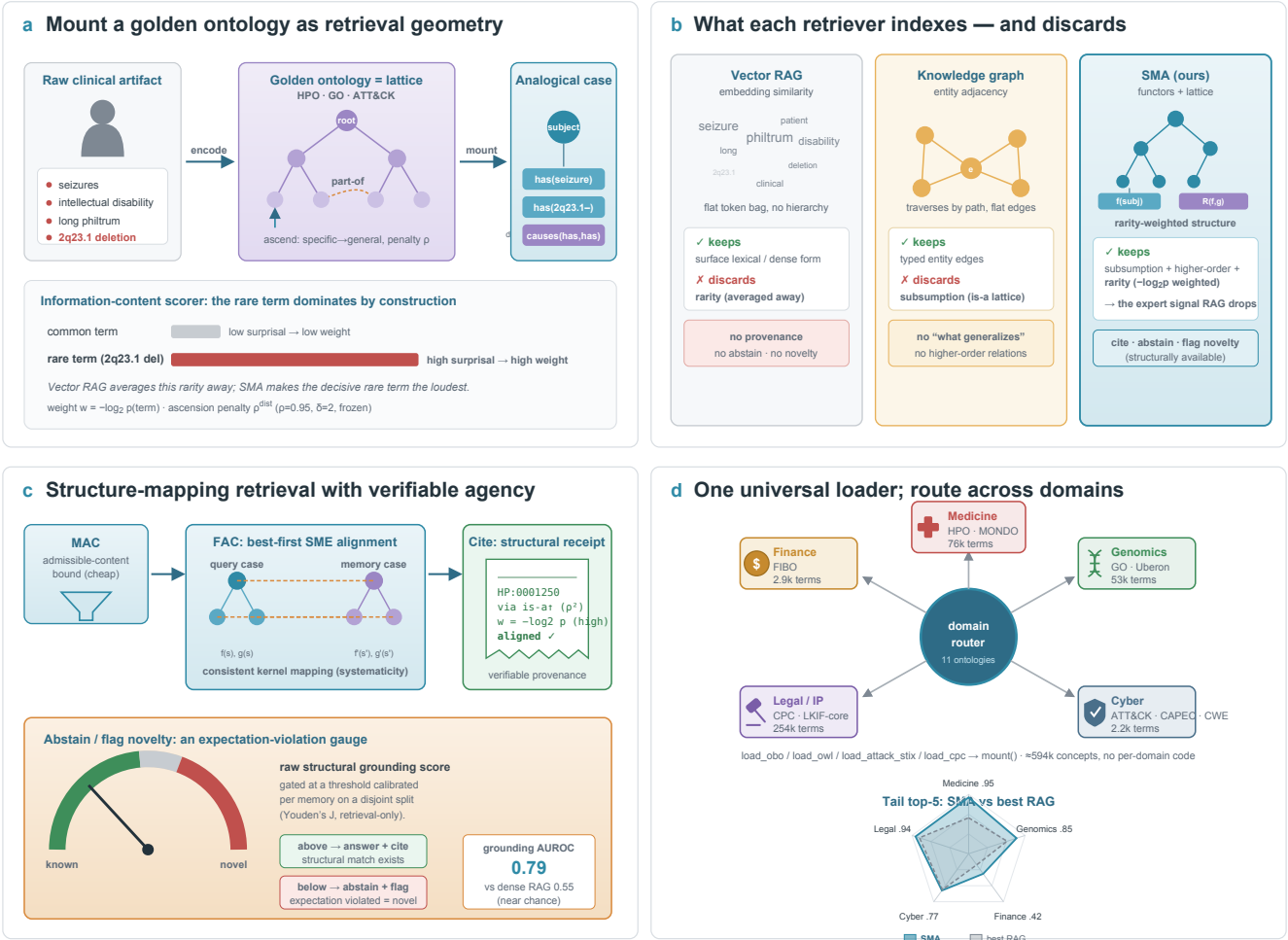


Figure 2 — Five-domain agentic results

MAIN (in manuscript) — tail top-5 by domain vs the enterprise RAG/KG gauntlet + effect-size forest

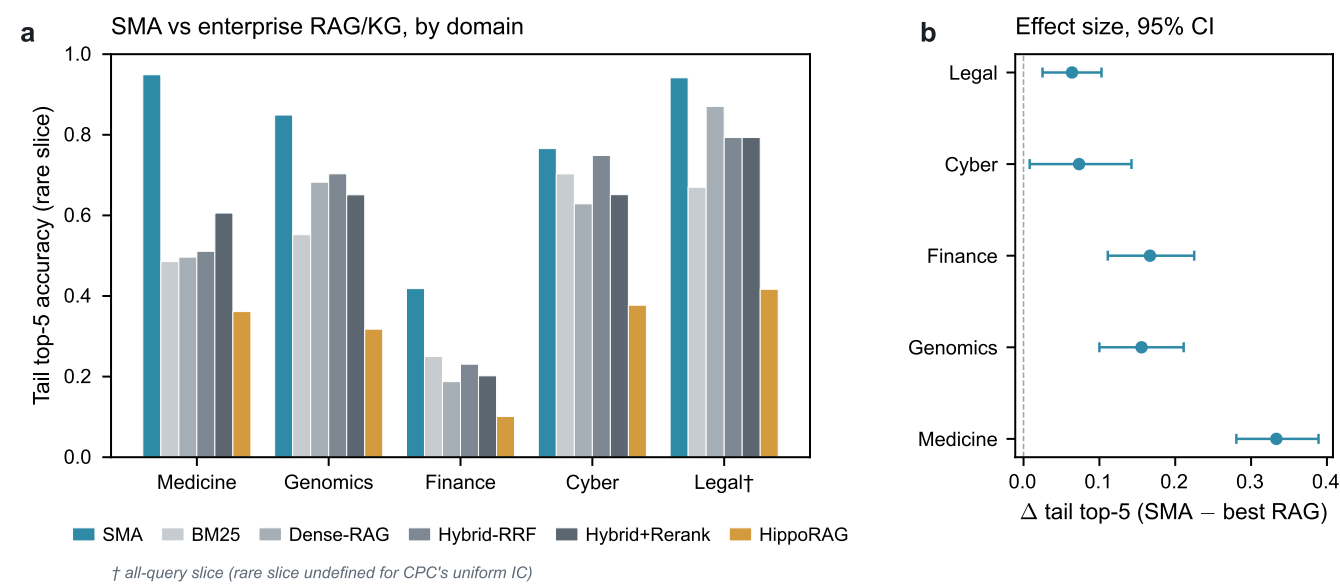


Figure 4 — Structural boundary

Alt — de-risk (ties the ontology oracle, beats RAG/KG) + the honest flat-tabular null

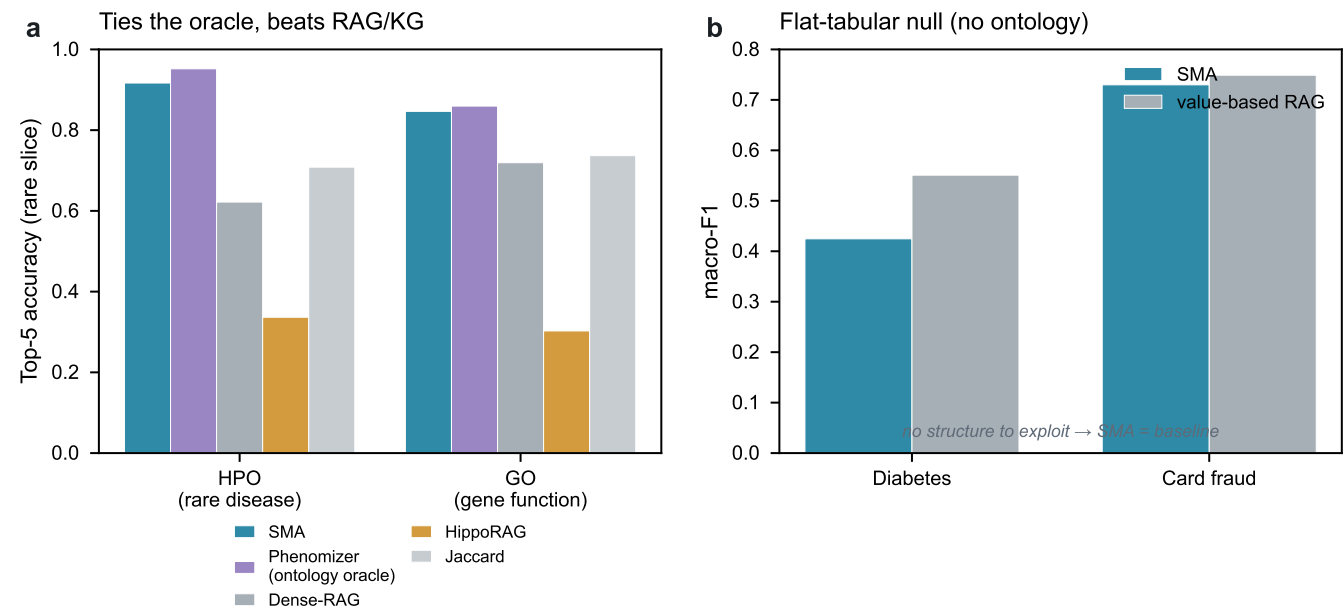


Figure 5 — Trustworthy-QA (verifiable specialist)

MAIN (in manuscript) — (a) trustworthy axes none/dense/SMA (b) grounding-score separation, known vs unknown

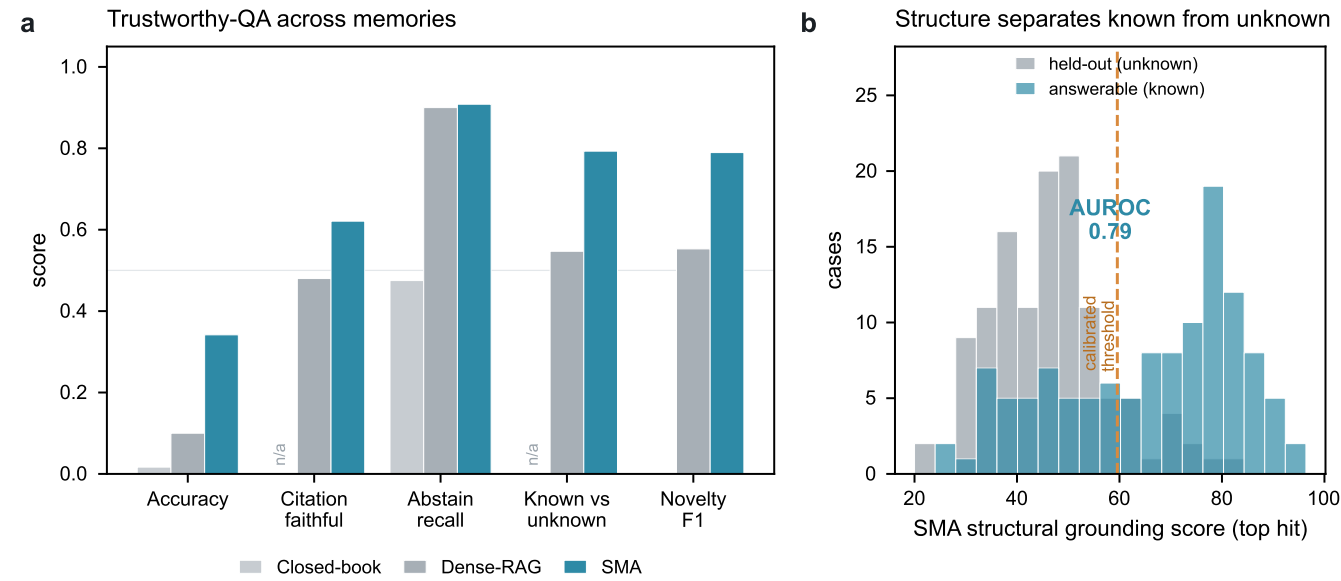
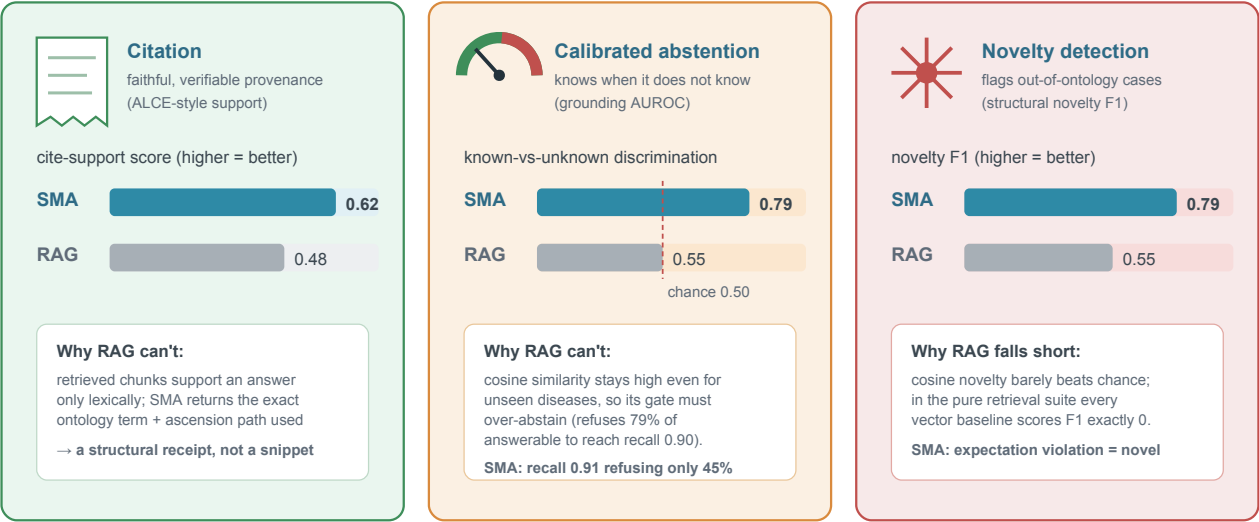


Figure 3 — Three capabilities (conceptual)

Alt — citation / grounding-AUROC / novelty-F1, SMA vs dense (manuscript currently uses Fig 5 for this)

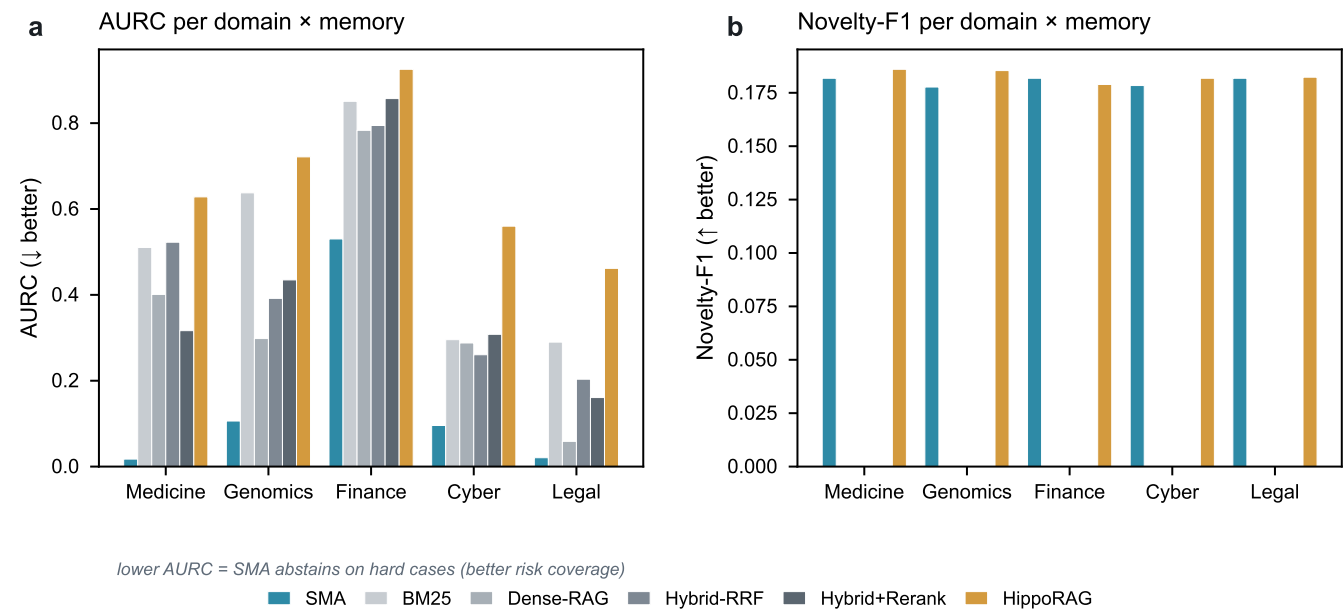
Three capability axes vector RAG and knowledge graphs structurally lack SMA dense RAG

Medicine LLM-QA (n = 120 answerable + 120 held-out); LLM and prompt fixed, memory swapped. SMA (teal) vs dense RAG (gray).



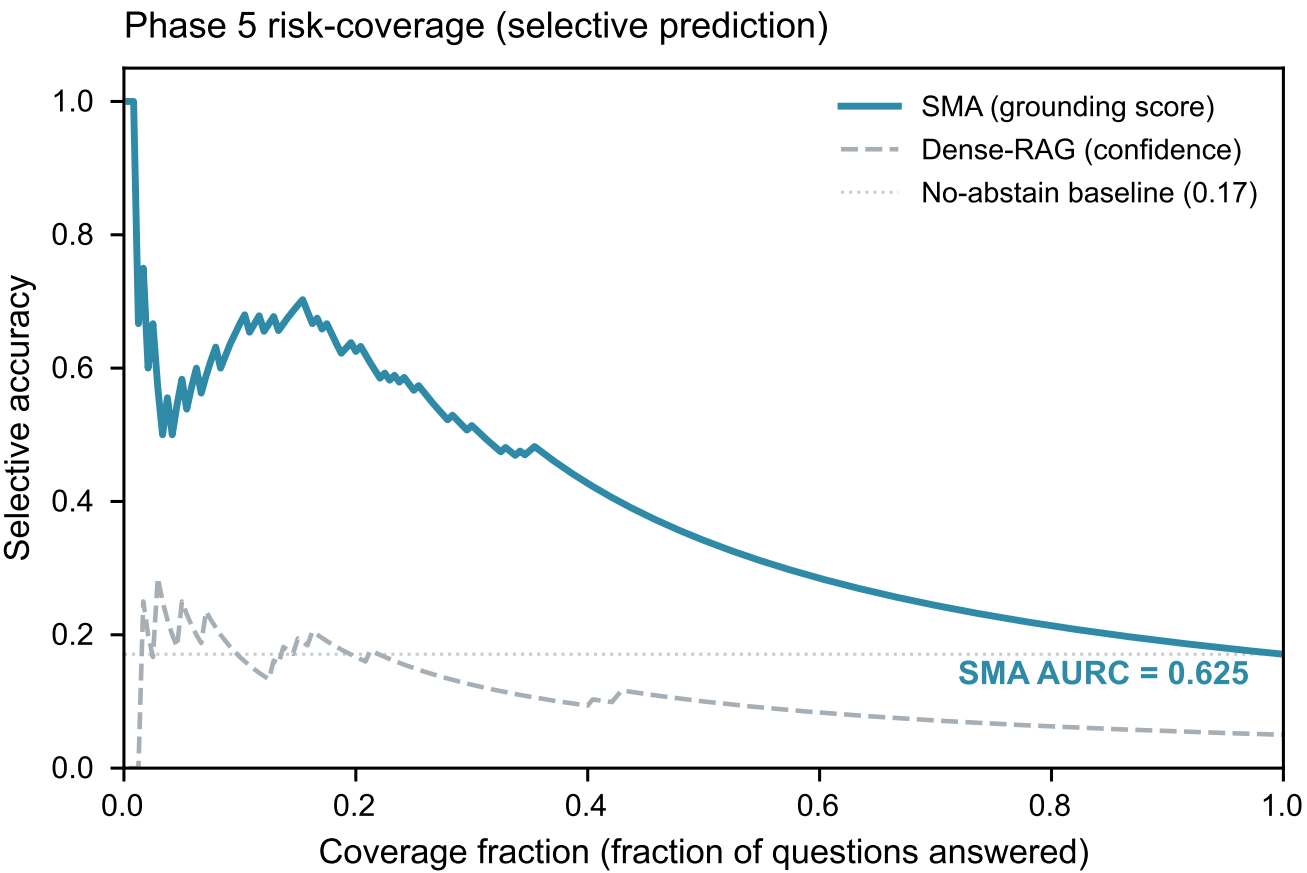
Extended Data Fig 1 — Per-domain AURC + novelty

ED (in supplement) — risk-coverage AURC and novelty-F1 across all six memories, five domains



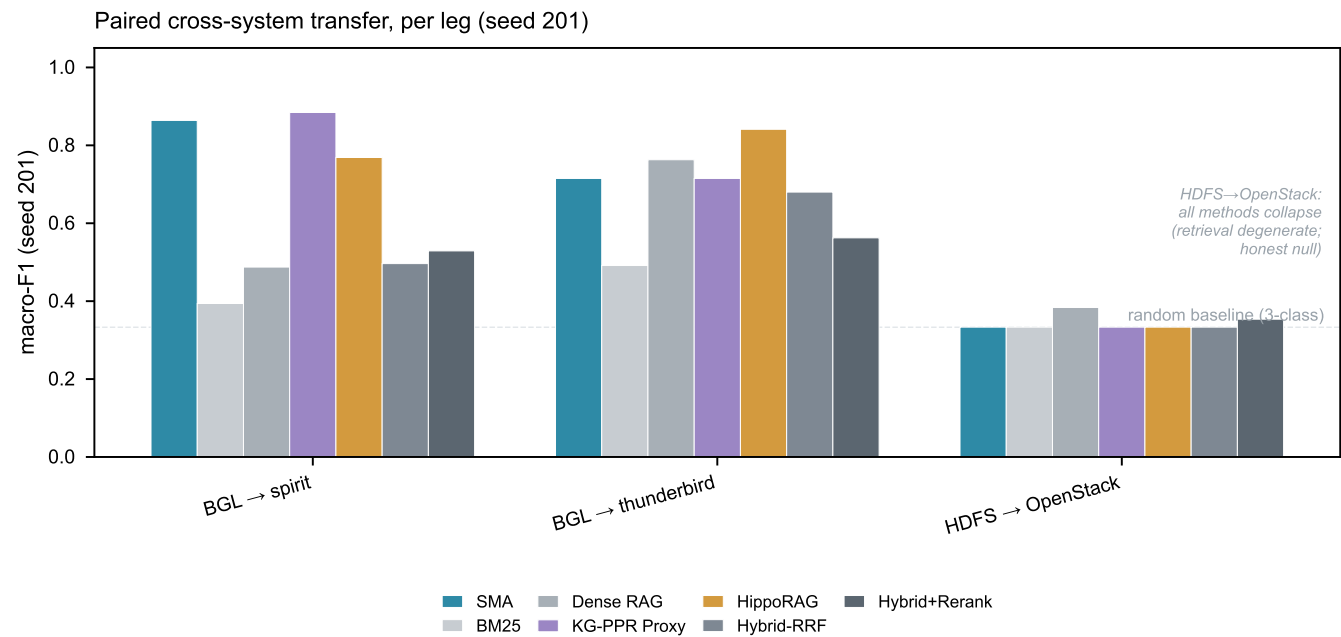
Extended Data Fig 2 — Phase 5 risk-coverage

ED (in supplement) — selective-prediction curve, SMA vs dense



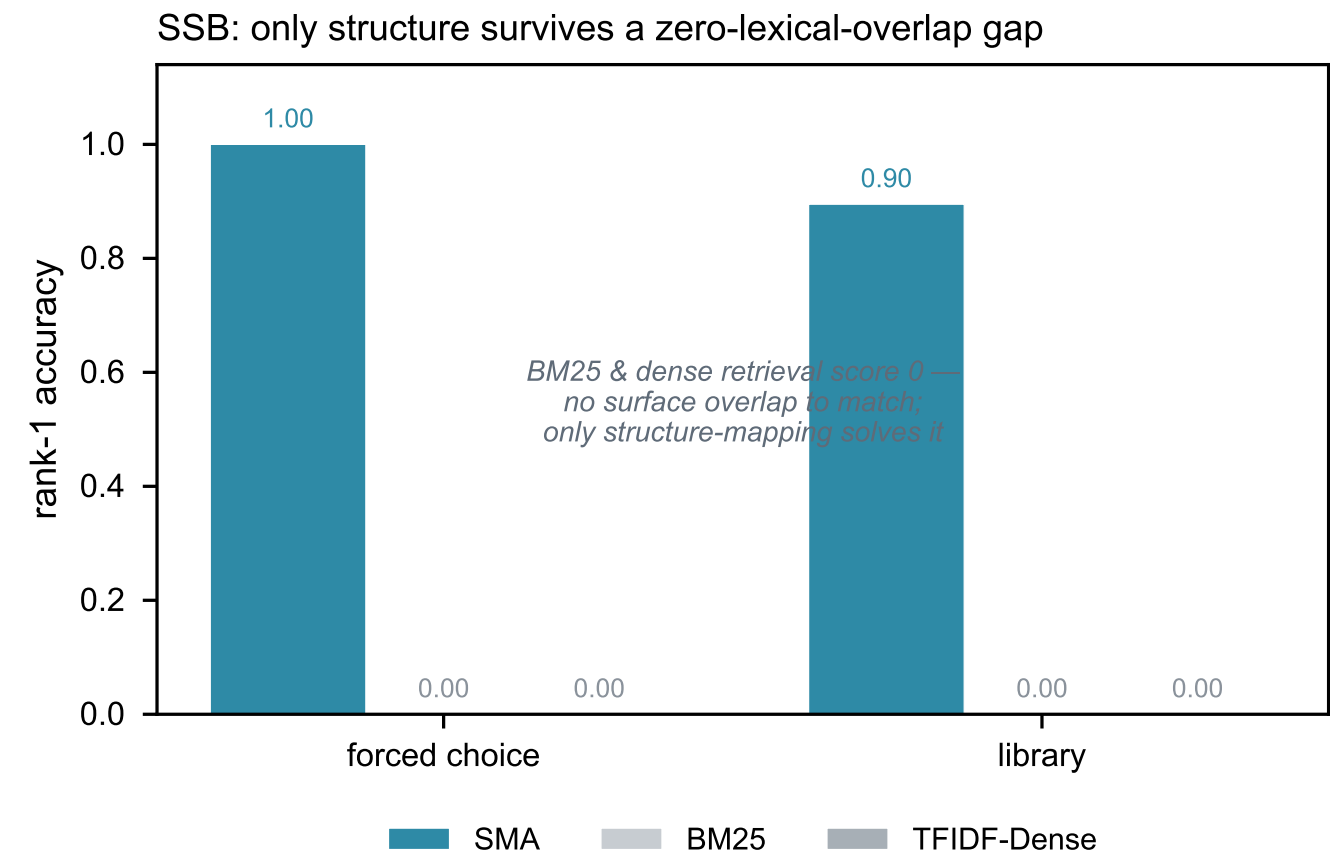
Extended Data Fig 3 — Paired cross-system transfer

ED (in supplement) — per-leg paired comparison (not leg-averaged)



Extended Data — SSB (structure vs surface)

ED (supplementary) — zero-lexical-overlap structural retrieval; lexical/dense baselines score 0



Extended Data — Frozen-dials ablation

ED (supplementary) — chosen config vs knocked-out knobs on the validation grid

